connection to the API and reject the request if it is different.

In both cases, the API can check that the client has authenticated with the same certificate by comparing the hash with the client certificate used to authenticate at the TLS layer. Listing 11.8 shows how to calculate the hash of the certificate, known as a thumbprint in the JOSE specifications, using the `java.security.MessageDigest` class that you used in chapter 4. The hash should be calculated over the full ASN.1 DER encoding of the certificate, which is what the `certificate.getEncoded()` method returns. Open the OAuth2TokenStore.java file in your editor and add the `thumbprint` method from the listing.

> **DEFINITION** A certificate ***thumbprint*** or ***fingerprint*** is a cryptographic hash of the encoded bytes of the certificate.

Listing 11.8 Calculating a certificate thumbprint

```
private byte[] thumbprint(X509Certificate certificate) {
    try {
        var sha256 = MessageDigest.getInstance("SHA-
256");     #A
        return sha256.digest(certificate.getEncoded());    #B
    } catch (Exception e) {
        throw new RuntimeException(e);
    }
}
```

#A Use a SHA-256 MessageDigest instance
#B Hash the bytes of the entire certificate